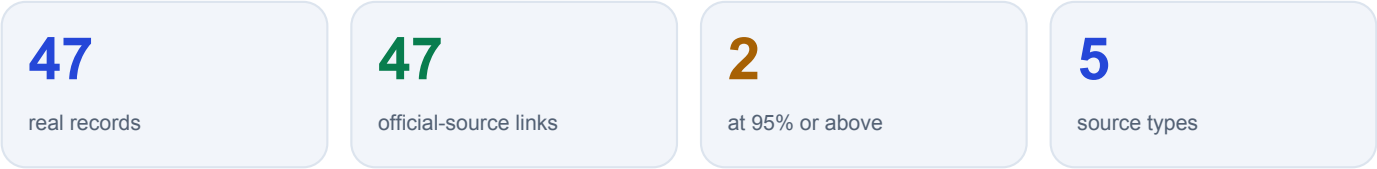


Scholarship Intelligence Crawler

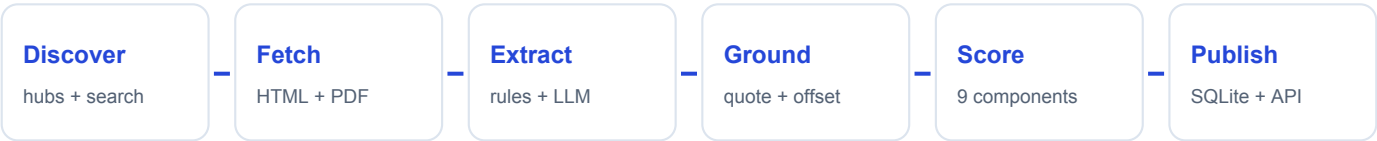
A working pipeline for Indian student funding: discover, crawl, extract, verify, score, store and update. Every retained field is tied to a fetched official-source page.

Live application

GitHub repository



HOW THE IMPLEMENTED PIPELINE WORKS



Discovery uses official hubs, topical search and provider-domain expansion. Aggregators supply leads only. Requests, BeautifulSoup, pypdf and optional Playwright build page snapshots; Gemini, Groq or Ollama are optional free extraction aids. FastAPI exposes the stored result.

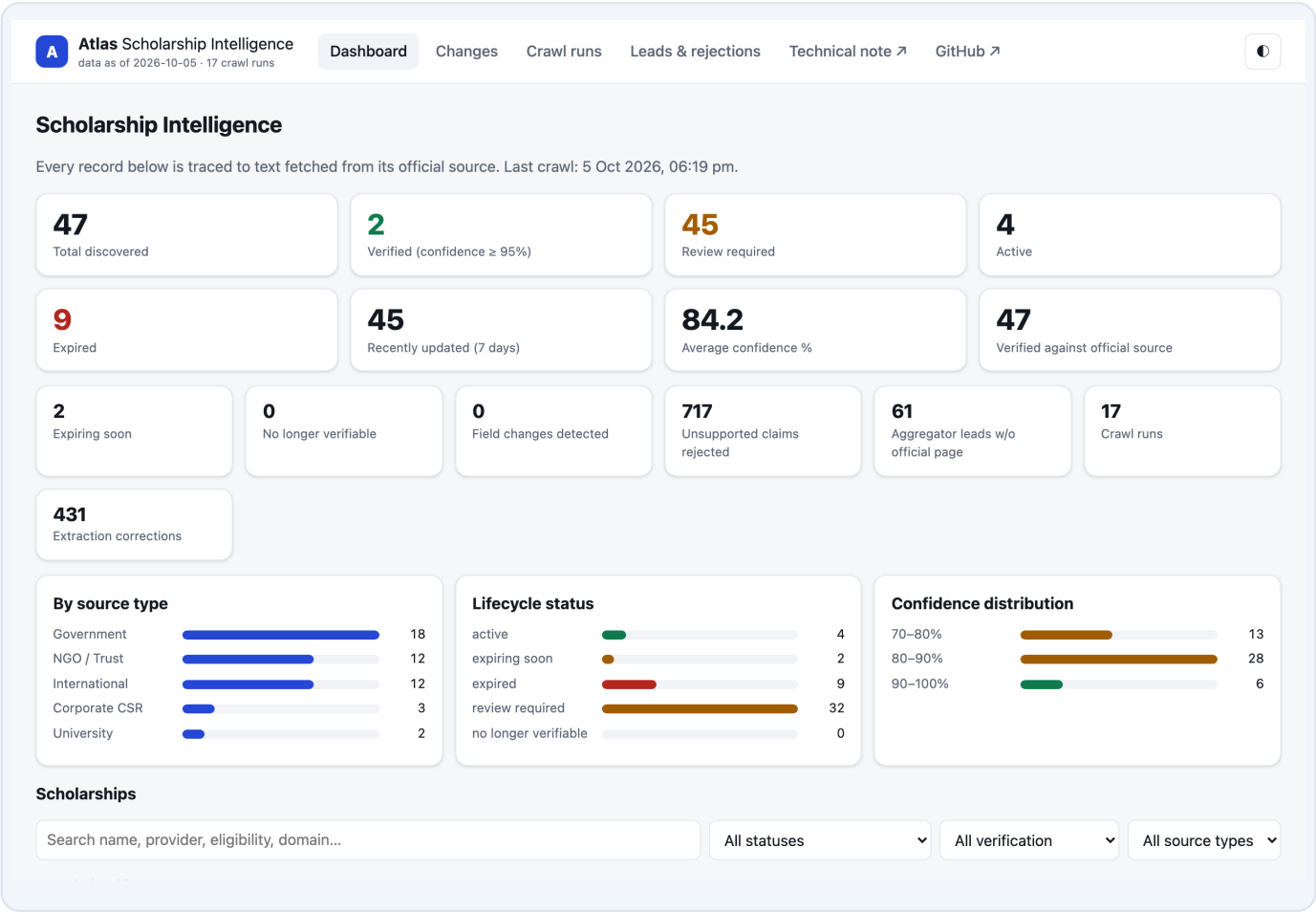


Figure 1. Dashboard captured from the submitted 47-record SQLite snapshot.
Observed result: 17 crawl runs | 546 current quoted field-evidence rows | 9 expired/stale records

From source page to traceable fact

A model can suggest a value, but the verification gate accepts it only when the stored official page contains the proposed passage and that passage supports the value.

Discovery and extraction

Ranked links and search leads resolve to provider, government or academic domains. HTML/PDF text is stored, then rules and optional LLMs propose normalized fields with verbatim quotes. A same-provider supporting notice can supply a deadline.

Grounding and rejection

The quote must match the page snapshot at recorded character offsets. Dates, amounts and categories must follow from that passage. Unsupported proposals are logged separately. Missing values stay Not specified.

A FIELD'S AUDIT PATH

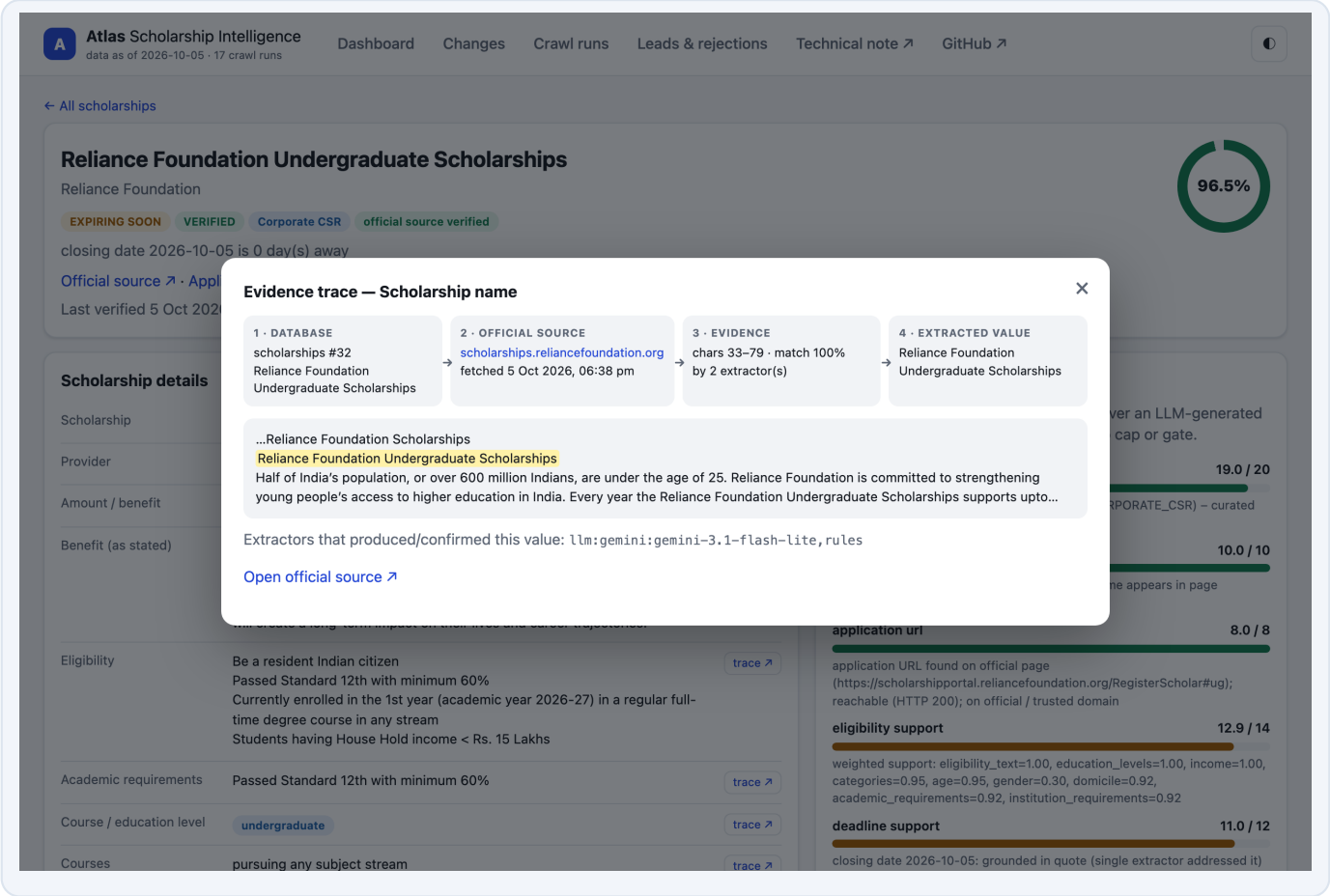


Figure 2. Real Reliance Foundation fact traced to an official-source passage.

Verification rule

The score is computed from evidence, not generated by a model. VERIFIED requires 95% or more and no blocking gate. Conflicting or absent critical evidence leaves the record REVIEW REQUIRED.

Measured confidence and repeatability

The score is a weighted sum of nine configured checks (maximum 100), followed by conflict penalties and hard caps. It is stored with a per-component explanation.

CONFIDENCE MODEL



Hard caps apply to non-official or unreadable pages, missing programme identity, insufficient independent extraction and unsupported critical fields. An expired deadline also reduces freshness.

REPEATED CRAWL AND CHANGE DECISION

Re-fetch known URL

Compare content hash

Ground changed fields

If source text is unchanged, an altered extraction is marked EXTRACTION CORRECTION. If the official page and grounded field change, retain old/new values, detection time, source and both evidence passages. Lifecycle tracks ACTIVE, EXPIRING SOON, EXPIRED, REVIEW REQUIRED and NO LONGER VERIFIABLE.

SUBMISSION AUDIT | ACTUAL SNAPSHOT

Records / official-source records	47 / 47	meets 20 / 15
Source types / expired or stale	5 / 9	meets 3 / 2
Records at 95%+	2	below 10
Observed official-source field changes	0	below 2

431 extractor/validator corrections were excluded from the observed source-change count. Real websites did not yield a documented field change across these runs; no source edits were manufactured.

DELIVERY AND REPRODUCTION

Read-only live app: Vercel serves the committed SQLite snapshot through FastAPI. GitHub Actions can crawl, export, checkpoint and commit future snapshots. The crawler uses free/open-source tools; model keys are optional and kept outside source control.

Inspect locally: `python -m scholarship_intel audit | python -m scholarship_intel list`
Run and serve: `python -m scholarship_intel run | python -m scholarship_intel serve --port 8010`
[edxso-scholarship-intelligence.vercel.app](#)

Limitations: missing official deadlines, incomplete eligibility, scanned PDFs and free-tier rate limits reduce confidence.